

REGIONAL GAS DISTRIBUTION MAINLINE - HFW/ERW QUALITY PLAN

Pipe Description: 20" (508 mm) x 9.53 mm, Grade X52N PSL 2, HFW / ERW (High-Frequency Welded) | Governing Standard: API Specification 5L 47th Edition / ISO 3183 PSL 2

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C <= 0.16%, P <= 0.020%, S <= 0.010%, CE_IIW <= 0.43	API 5L Tablo 5	H/W
2	Product Chemical Analysis	Two analyses per test unit	Pipe body coupon	ISO 14284	C <= 0.16%, P <= 0.020%, S <= 0.010%, CE_IIW <= 0.43	API 5L Tablo 5	W
3	Pipe Body Tensile Test	Once per test unit (lot <= 100 pipes)	Transverse body strap	ASTM A370	Rt0.5: 360-530 MPa, Rm: 460-760 MPa, Af >= 23%, Y/T <= 0.93	API 5L Tablo 7	H/W
4	Weld Seam Tensile Test	Once per test unit	Transverse weld coupon	ASTM A370	Rm >= 460 MPa across electric resistance weld seam	API 5L Tablo 7	W
5	Flattening Test (Düzlemtirme Testi)	Crop ends of each coil (first and last pipe of lot)	Full pipe section ring	API 5L 9.10.2	Step 1: Weld at 90° to 67% D (no opening); Step 2: to 33% D (no cracks)	API 5L 9.10.2 & Tablo 18	H/W
6	CVN Charpy Impact Test (Body & Weld) at 0 °C	1 set body + 1 set weld per test unit	Body & weld Charpy V	ASTM A370	Body: Min Avg 40 J, Min Ind 30 J; Weld: Min Avg 27 J, Min Ind 20 J (0 °C)	API 5L Tablo 8	H/W
7	Drop Weight Tear Test (DWTT)	Once per heat at 0 °C (D=508 mm threshold)	Transverse DWTT coupon	API RP 5L3	At 0 °C: Average shear area >= 85%	API 5L Tablo 18	H/W
8	Mill Hydrostatic Pressure Test	Each pipe (100% all pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 104 bar (90% SMYS), Duration: MINIMUM 10 SECONDS	API 5L 10.2.6.2	H/W
9	Electric Weld Seam UT Inspection	100% full length of HFW weld seam	HFW weld line	ISO 10893-10	100% automated Ultrasonic Testing of weld seam after seam heat treatment	API 5L Ek E	H/W
10	Pipe Ends UT Laminar Testing	100% of pipe ends	Both pipe ends	API 5L Annex E.8	100 mm band on pipe ends; no defect > 6.0 mm	API 5L Ek E.8	W
11	Dimensional & Wall Thickness Verification	100% of all pipes	Pipe body and ends	API 5L 10.2.8	Diameter: +-0.50%; Ovality: per Table 10; Wall thickness: -10.0% / +15.0%	API 5L Tablo 10/11	W
12	Visual Surface & Magnetism Inspection	100% internal and external	Full pipe surface	API 5L 10.2.7	No defect > 12.5% t; Residual magnetism <= 3.0 mT (30 Gauss)	API 5L 9.12 & 10.2.8.5	W